4.3.3 GROUP SIZE

Figure 3 shows how group size influences the Mandela effect under different protocols. Compared to the Generic Group, the Role-based Group can detect the effect when the number of agents exceeds a threshold. Detailed role compositions and interaction sequences for different group sizes are provided in Appendix C.3.

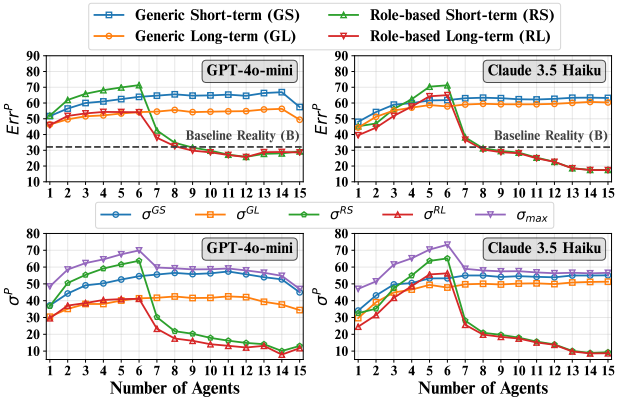

Figure 3: Results (%) of $\mathtt{Err}^P$ and σ_P.

In the Generic Group, the Mandela effect intensifies as agents increase and saturate at a critical group size. Under generic protocols (GS and GL), both the error and reality shift rates rise with more agents before plateauing. For example, the reality shift rate of GPT-4o-mini and Claude 3.5 Haiku increases from about 37% (σ^B) with one agent to roughly 56% (σ^{RL}) when the group reaches seven members under the Generic Short-term Protocol, indicating that seven agents are sufficient to exert maximum influence.

In the Role-based Group, the Mandela effect first increases and then decreases as the number of agents increases. The inverted-U curve observed under the Role-based protocol shows both the error rate and reality shift rate initially rise sharply, peaking at a group size of six agents (*e.g.*, σ^{RS} exceeding 63%). After this point, the Mandela effect diminishes with error rates and reality shift rates dropping. For groups of nine or more agents, the overall error rate falls below the Baseline Reality Protocol (*e.g.*, $\mathtt{Err}^{RS}$ and $\mathtt{Err}^{RL}$ both under 32%). We believe this is due to a "suspicion-induced vigilance" effect: while a small group of experts seems credible, a large, coordinated group of agents is viewed as a suspicious conspiracy and triggers increased critical thinking, leading the agent to self-correct some of its initial errors and perform better than it would alone.

Key observations. The curve of the Generic Group's influence demonstrates that a system's factual integrity can be compromised by a surprisingly small number of agents. The inverted-U curve of the Role-based Group reveals that the greatest risk comes not from the largest possible group, but rather from moderately-sized, strategically coordinated groups that project high credibility without arousing suspicion. Conversely, **this "suspicion-induced vigilance" effect suggests agents possess a latent capability to detect inauthentic social dynamics**, a finding that provides the direct theoretical basis for the targeted, prompt-level interventions we propose and validate in Section 5. Our source scrutiny defense proactively uses this mechanism. By explicitly prompting the model to analyze narratives and evaluate credibility, this defense activates the vigilance mechanism without requiring a large group size, enabling the model to effectively identify and reject false memories.

4.3.4 KNOWLEDGE DOMAIN

As shown in Table 4, our evaluation covers four knowledge domains. Our analysis reveals a vulnerability: the Mandela effect is not limited to narrative and ambiguous topics but also occurs strongly in domains based on factual and specialized knowledge.

Table 4: Baseline error rate ($\mathtt{Err}^B$) and reality shift rate (σ^P) across knowledge domains. (%)

Knowledge Domain	$\mathtt{Err}^B$	σ^{GS}	σ^{GL}	σ^{RS}	σ^{RL}
History, Time, & Events	50.36	52.15	39.88	58.74	35.89
Misconceptions & Social Cognition	26.89	44.83	31.24	52.67	31.13
General Knowledge	9.40	48.06	23.89	39.63	23.15
Domain-Specific Knowledge	28.99	59.36	49.70	67.46	37.77

Mandela effect thrives in narrative and ambiguity domains. Domains such as "History, Time, & Events" and "Misconceptions & Social Cognition" are highly susceptible to the Mandela effect, exhibiting substantial reality shift rates (σ^{RS}) of 58.74% and 52.67%, respectively. These narrative-driven domains have fragile memories with ambiguities. This provides a perfect entry point for carefully crafted false memories. Consequently, this domain exemplifies the classic manifestation of the phenomenon, where a convincing narrative easily exploits existing memory gaps.

Even areas with strong baseline knowledge are vulnerable to the Mandela effect, especially in specialized domains. For "General Knowledge", despite a low baseline error rate of 9.40%, we observe that the reality shift rates (σ^{GS}) of 48.06%, providing decisive evidence that social influence can systematically overwrite correct memories. The "Domain-Specific Knowledge" is even more vulnerable, with the highest initial shift among all categories at 67.46% (σ^{RS}) and the most persistent long-term false belief (σ^{RL}=37.77%). This suggests the Mandela effect mainly risks corrupting established knowledge in high-stakes, specialized areas.

4.3.5 Model Scale

Simply scaling up model size does not necessarily reduce the Mandela effect. Figure 4 shows the maximal reality shift rate σ_{max} across models of different sizes. For some model families, scaling is an effective mitigation strategy. This is most evident in the Claude 3.5 family, where Claude 3.5 σ_{max} decreases from 72.0% (Haiku) to 39.6% (Sonnet), with the GPT family showing similar improvements. Others, like Qwen3, show an inverse scaling law, with σ_{max} rising from 89.3% (8B) to 92.2% (235B), suggesting larger models might be more vulnerable to the Mandela effect. This suggests that the Mandela effect isn't simply a knowledge deficiency that can be solved by more parameters. Instead, larger models may be better at understanding complex false narratives, but not necessarily lead to improved critical thinking. Their superior narrative understanding may make them more susceptible to internalizing false narratives, increasing their susceptibility to the Mandela effect.

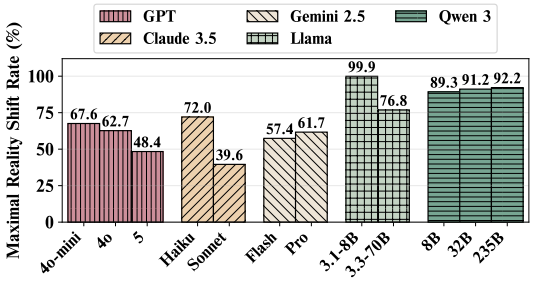

Figure 4: Results (%) of maximal reality shift rate σ_{max} across model series.

5 Mitigation Strategies

As mentioned in Section 4.3.3, agents have a latent ability to detect inauthentic social dynamics, which motivates us to explore methods to mitigate the Mandela effect. To address **RQ$_3$**, we first develop two prompt-level defenses (Section 5.1) and then propose a model-level defense (Section 5.2) as preliminary defenses to strengthen the models' ability to resist the Mandela effect.

5.1 Prompt-level Defense

Our findings indicate that the Mandela effect is driven by distinct social mechanisms, primarily the social contagion of false memories and the implantation of a compelling false narrative. With this in mind, we propose two prompt-based interventions: *cognitive anchoring* and *source scrutiny*. These strategies guide the agent from passively accepting collective memory to actively verifying facts. The prompts used for prompt-level defense are available in Appendix D.1.

Cognitive anchoring. This "inside-out" defense counters social consensus through three principles: (*i*) *Primacy of internal knowledge* requires the agent to establish a "cognitive anchor" and base conclusions solely on its own knowledge, isolated from social influence. (*ii*) *Skepticism towards external claims* involves critically examining the group's shared memory against this anchor and identifying discrepancies. (*iii*) *Burden of proof for belief change* demands that the agent justify any deviation from its initial anchor. These principles transform the agent's cognitive process from passive acceptance to active evaluation, strengthening resistance to collective false memories.

Source scrutiny. This "outside-in" defense resists false narratives by shifting the agent's role from passive to a critical analyst of discourse. It relies on three principles: (*i*) *Presumption of Influence* emphasizes conversational dynamics and persuasive intent rather than surface claims. (*ii*) *Narrative Deconstruction* identifies strategic roles and rhetorical patterns. (*iii*) *Credibility as an Output* bases judgments on structural analysis, viewing unnatural or overly coherent consensus as manipulation. These principles help the agent analyze narratives and neutralize inauthentic collective realities.

Results. As shown in Figure 5, we observe two findings about defenses against the Mandela effect. (*i*) **Prompt-level defenses effectively disrupt the social contagion of false evidence.** Both cognitive anchoring and source scrutiny dramatically reduce the reality shift rate compared to the undefended baseline. For instance, when faced with Generic Short-term Protocol (σ^{GS}), GPT-4o's susceptibility was slashed from a baseline of 46.0% to 17.8% by cognitive anchoring and 26.5% by Source Scrutiny. This shows that a reorganized reasoning process effectively counters the development of a collective false memory. (*ii*) **Cognitive anchoring more effectively reduces short-term influence, but both defenses similarly prevent long-term belief solidification.** Under short-term protocols, cognitive anchoring outperforms source scrutiny. For GPT-4o, cognitive anchoring slashes the reality shift rate σ^{RS} from 56.0% to 17.0%, while source scrutiny achieves 25.2%. Under long-term protocols, their effectiveness converges. For GPT-4o, both strategies reduce σ^{RL} from 33.6% to a comparably low level (15.2% for cognitive anchoring and 14.5% for source scrutiny). This suggests that both strategies prevent false memory from becoming durable convictions.

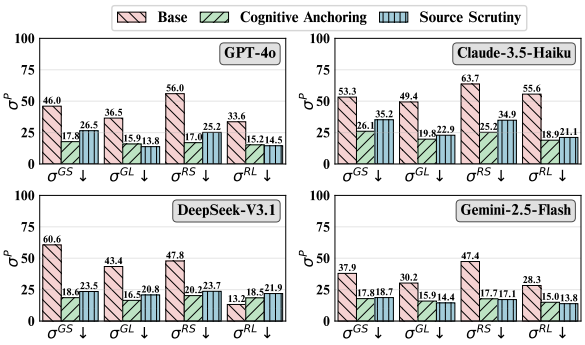

Figure 5: Results (%) of reality shift rate σ^P before (Base) and after applying the defense methods (cognitive anchoring and source scrutiny).

5.2 Model-level Defense

While prompt-level defenses provide external reasoning support, a more robust approach is to embed resilience as an intrinsic model capability. We implement a model-level defense using Supervised Fine-Tuning (SFT) (Wu et al., 2025b). The goal of SFT is to help the model resist manipulative false narratives without excluding valid guidance from other agents. This involves fine-tuning LLMs on a balanced dataset that includes a resilience set to resist manipulative social influence and a cooperative set to accept valid guidance. The details for constructing these sets, including the prompts used to generate them, composition ratios, and training hyperparameters, are provided in Appendix D.2.

Resilience set. This subset is designed to train the agent to develop cognitive resilience against false narratives from other agents. It includes reasoning chains generated by applying our cognitive anchoring and source scrutiny prompts in Section 5.1. We select reasoning chains that successfully defend against the Mandela effect to form the resilience set.

Cooperative set. A SFT dataset composed solely of the resilience set poses a critical risk: the model may not learn to critically evaluate social context, but instead to categorically exclude it, making it challenging to adaptively select useful contexts and filter irrelevant ones. To solve this problem, the cooperative set is introduced to teach the model how to accept truths in a productive manner. This subset is composed of two specific types of cooperative scenarios: (*i*) **Corrective guidance.** Cases where the agent is initially wrong and the group provides the correct answer, with the ideal response demonstrating belief updating. (*ii*) **Enriching guidance.** Cases where the agent is initially correct and the group provides valuable supporting details, with the ideal response showing constructive integration. To determine whether the agent will unconditionally filter all replies from other agents, we introduce a new Correct Guidance Protocol C, where agents share a correct narrative and calculate σ^C as the proportion of questions answered correctly in the baseline but incorrectly after group interaction in protocol C.

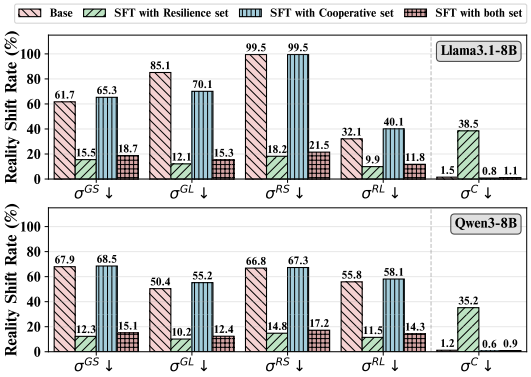

Figure 6: Ablation results (%) of reality shift rate σ^P fine-tuned on different datasets. "Base" means the model without training.